

5.2 Some Popular Transformer Models: BERT, GPT, and BART

5.2.3 BERT: Bidirectional Encoder Representations from Transformers

BERT



(Bidirectional Encoder Representations from Transformers)

Paper: Devlin J, Chang MW, Lee K, Toutanova K. BERT: Pre-training of deep bidirectional transformers for language understanding.

<https://arxiv.org/abs/1810.04805>

(Google Research, 2018)

- multi-layer bidirectional transformer encoder
- architecture almost identical to original transformer & GPT, except
 - bidirectional masking (known as "Cloze" task, Taylor 1953*)
 - next sentence prediction as additional pre-training task

* Wilson L Taylor. 1953. Cloze procedure: A new tool for measuring readability. *Journalism Bulletin*, 30(4):415–433.

BERT Inputs

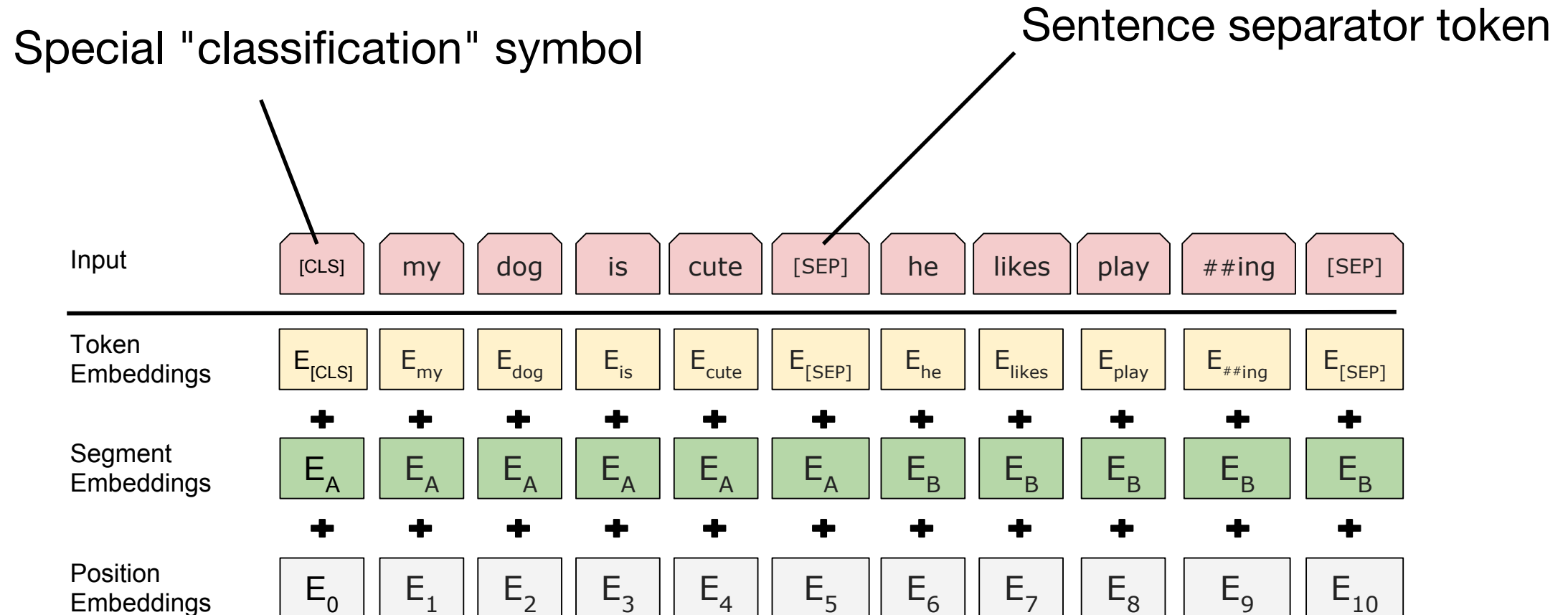


Figure 2: BERT input representation. The input embeddings are the sum of the token embeddings, the segmentation embeddings and the position embeddings.

Token embeddings are WordPiece embeddings* with vocabulary size of 30,000

* Wu Y, Schuster M, Chen Z, Le QV, Norouzi M, Macherey W, Krikun M, Cao Y, Gao Q, Macherey K, Klingner J. Google's neural machine translation system: Bridging the gap between human and machine translation. arXiv preprint arXiv:1609.08144. 2016 <https://arxiv.org/abs/1609.08144>

BERT Pre-Training Tasks

Pre-training datasets

- BookCorpus (800 million words)
- Wikipedia (2500 million words)

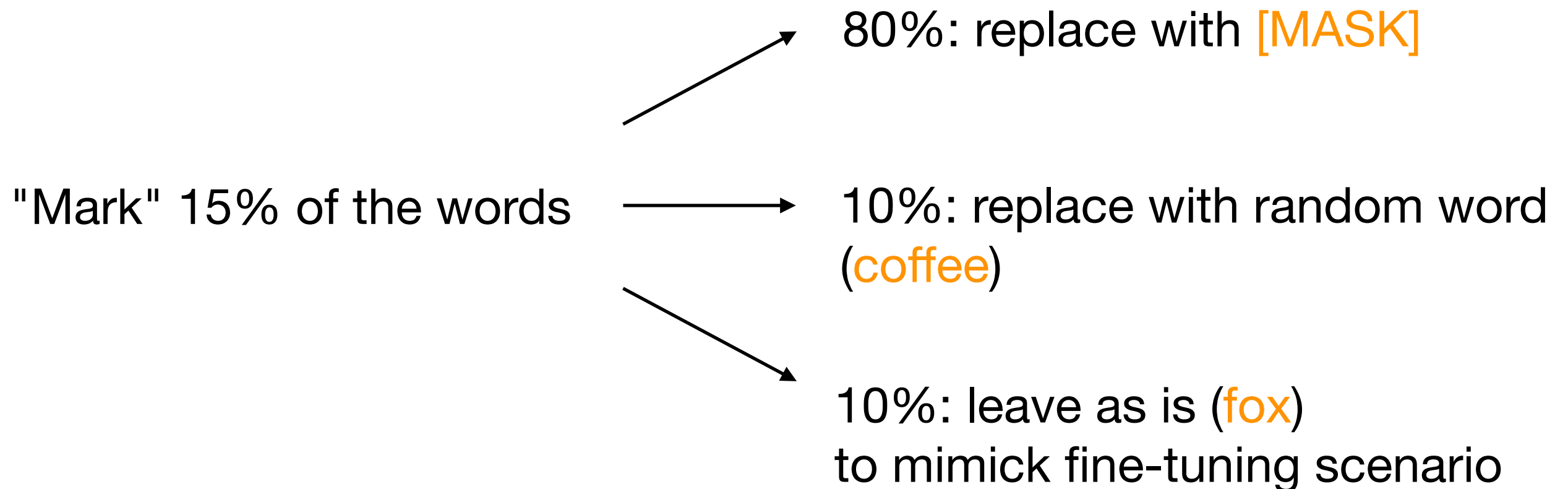
Pre-training tasks

- Masked language model ("Cloze")
- Next sentence prediction

BERT Pre-Training Task #1

Masked Language Model

Input sentence: A quick brown fox jumps over the lazy dog



BERT Pre-Training Task #1

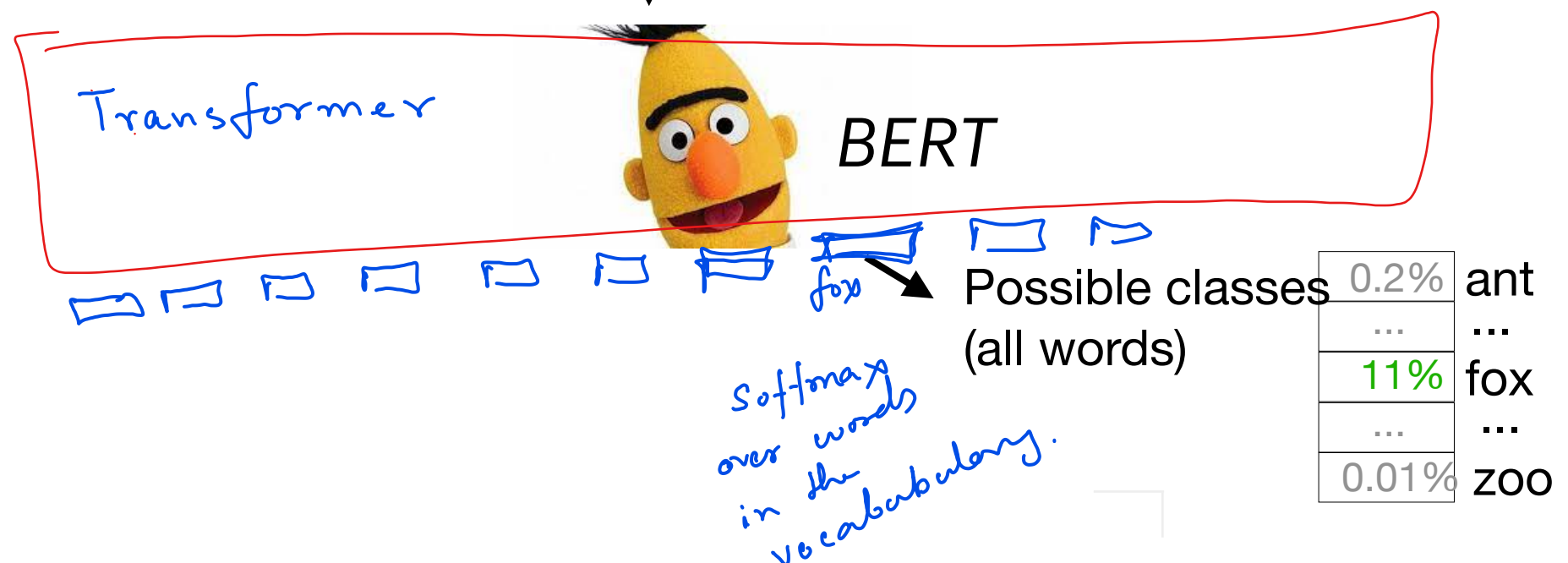
Masked Language Model

Input sentence:

A quick brown fox jumps over the lazy dog

Randomly masked:

A quick brown [MASK] jumps over the lazy dog



BERT Pre-Training Task #2

Next Sentence Prediction

Balanced binary classification task (50% IsNext, 50% NotNext)

Input = [CLS] the man went to [MASK] store [SEP] he bought a gallon [MASK] milk [SEP]

Label = IsNext

Input = [CLS] the man [MASK] to the store [SEP] penguin [MASK] are flight ##less birds [SEP]

Label = NotNext

↗

BERT Pre-Training & Downstream Tasks

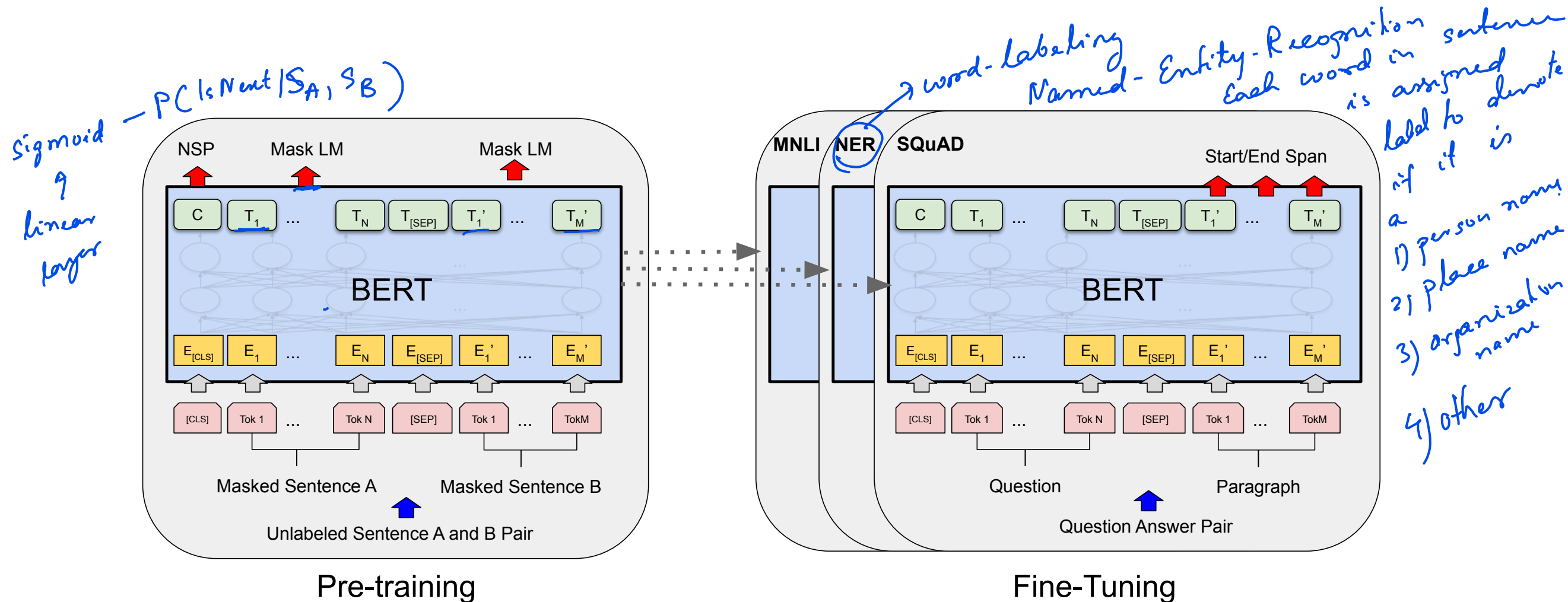


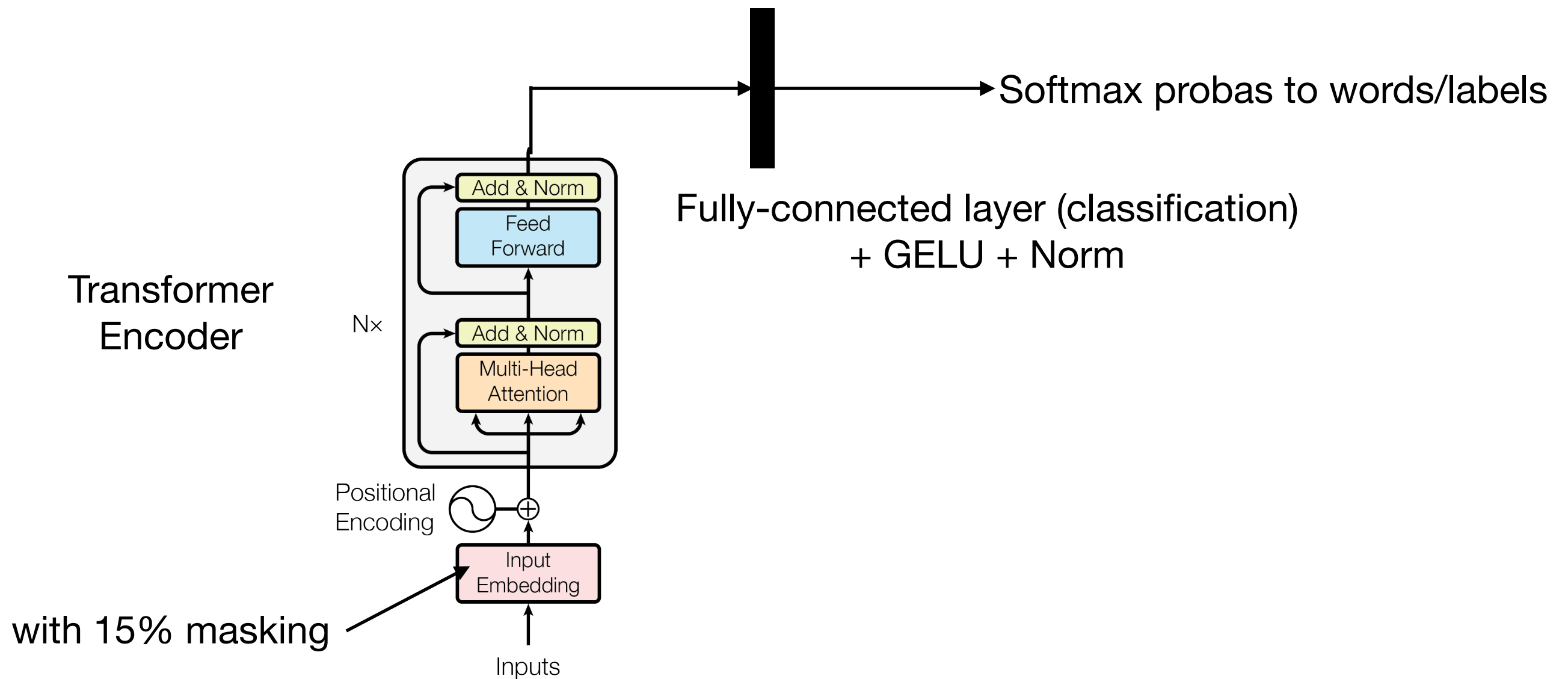
Figure 1: Overall pre-training and fine-tuning procedures for BERT. Apart from output layers, the same architectures are used in both pre-training and fine-tuning. The same pre-trained model parameters are used to initialize models for different down-stream tasks. During fine-tuning, all parameters are fine-tuned. [CLS] is a special symbol added in front of every input example, and [SEP] is a special separator token (e.g. separating questions/answers).

Transformer Training Approach

1. **Pre-training** on large unlabeled datasets
(self-supervised learning)
2. **Training for downstream-tasks** on labeled data
(supervised learning)
 - a) fine-tuning approach
 - b) feature-based approach
(nowadays also called "fine-tuning")

BERT Pre-Training & Fine-Tuning Approach

- Add classification layer
- Train end-to-end on labeled dataset for downstream task (update ALL parameters)



BERT vs GPT-v1 Performance

System	MNLI-(m/mm) 392k	QQP 363k	QNLI 108k	SST-2 67k	CoLA 8.5k	STS-B 5.7k	MRPC 3.5k	RTE 2.5k	Average -
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.8	90.4	36.0	73.3	84.9	56.8	71.0
OpenAI GPT	82.1/81.4	70.3	87.4	91.3	45.4	80.0	82.3	56.0	75.1
BERT _{BASE}	84.6/83.4	71.2	90.5	93.5	52.1	85.8	88.9	66.4	79.6
BERT _{LARGE}	86.7/85.9	72.1	92.7	94.9	60.5	86.5	89.3	70.1	82.1

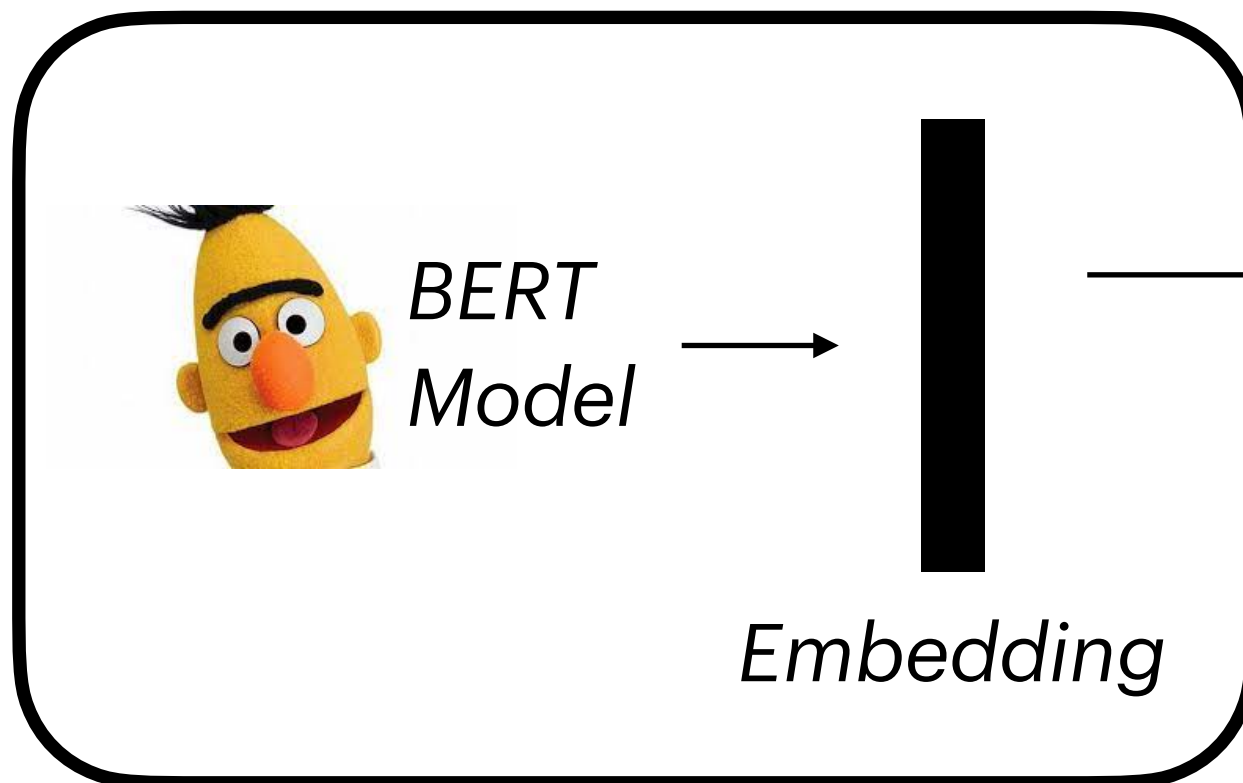
Table 1: GLUE Test results, scored by the evaluation server (<https://gluebenchmark.com/leaderboard>). The number below each task denotes the number of training examples. The “Average” column is slightly different than the official GLUE score, since we exclude the problematic WNLI set.⁸ BERT and OpenAI GPT are single-model, single task. F1 scores are reported for QQP and MRPC, Spearman correlations are reported for STS-B, and accuracy scores are reported for the other tasks. We exclude entries that use BERT as one of their components.

BERT Pre-Training & Feature-based Training

- Keep BERT frozen after pre-training
- Create BERT embeddings for labeled dataset for downstream task and train new model on these embeddings
(in original paper, 2-layer biLSTM on embeddings from concatenated last 4 layers performed best)

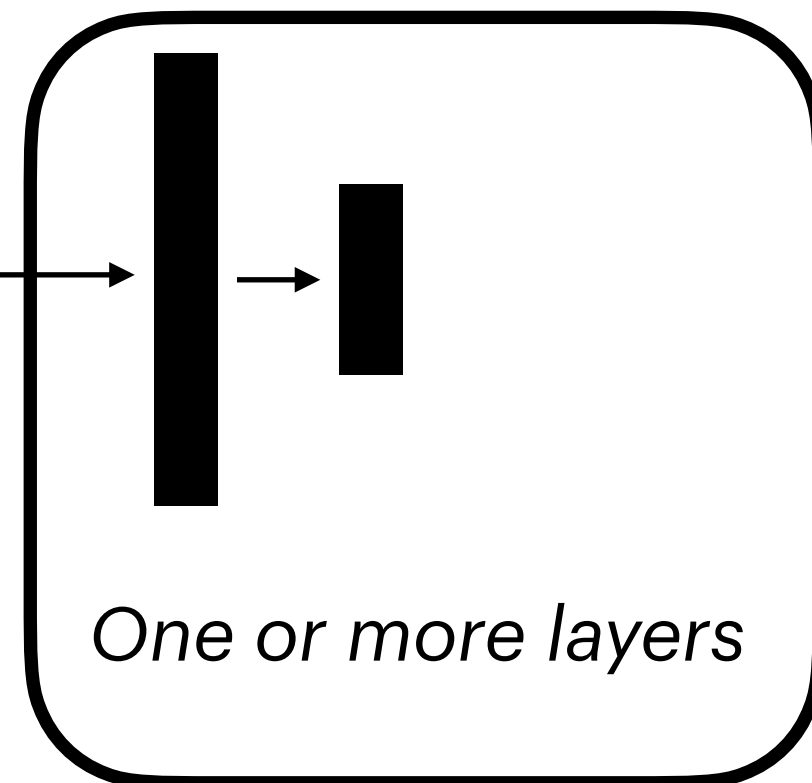
1) Download BERT

pre-trained on large corpus
(in self-supervised fashion)



2) Feature-based training ("fine-tuning")

on target task
(supervised learning)



System	Dev F1	Test F1
ELMo (Peters et al., 2018a)	95.7	92.2
CVT (Clark et al., 2018)	-	92.6
CSE (Akbik et al., 2018)	-	93.1
Fine-tuning approach		
BERT _{LARGE}	96.6	92.8
BERT _{BASE}	96.4	92.4
Feature-based approach (BERT _{BASE})		
Embeddings	91.0	-
Second-to-Last Hidden	95.6	-
Last Hidden	94.9	-
Weighted Sum Last Four Hidden	95.9	-
Concat Last Four Hidden	96.1	-
Weighted Sum All 12 Layers	95.5	-

Table 7: CoNLL-2003 Named Entity Recognition results. Hyperparameters were selected using the Dev set. The reported Dev and Test scores are averaged over 5 random restarts using those hyperparameters.